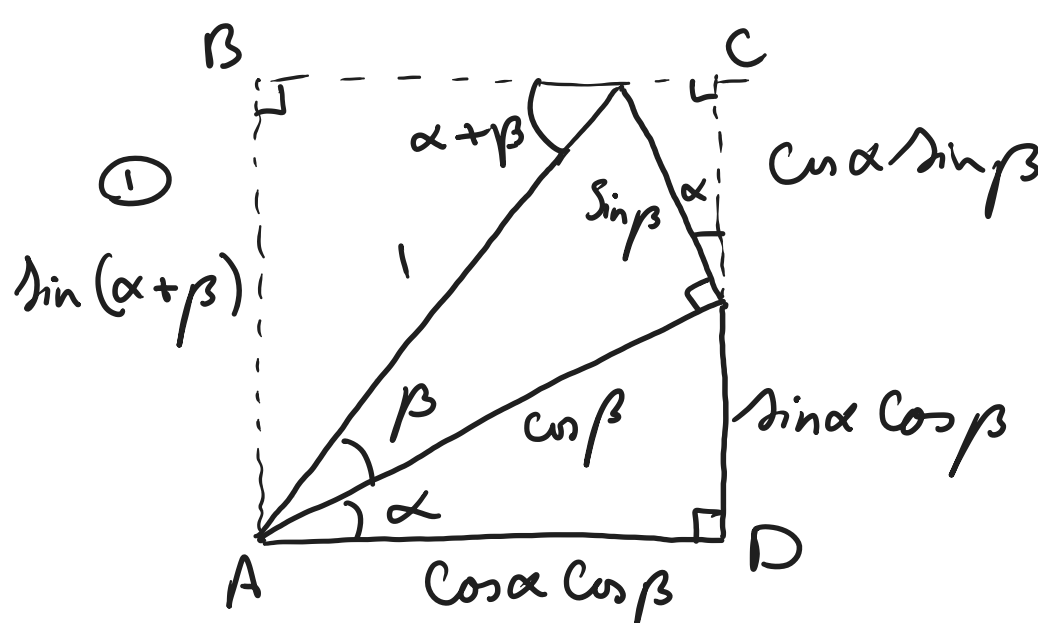


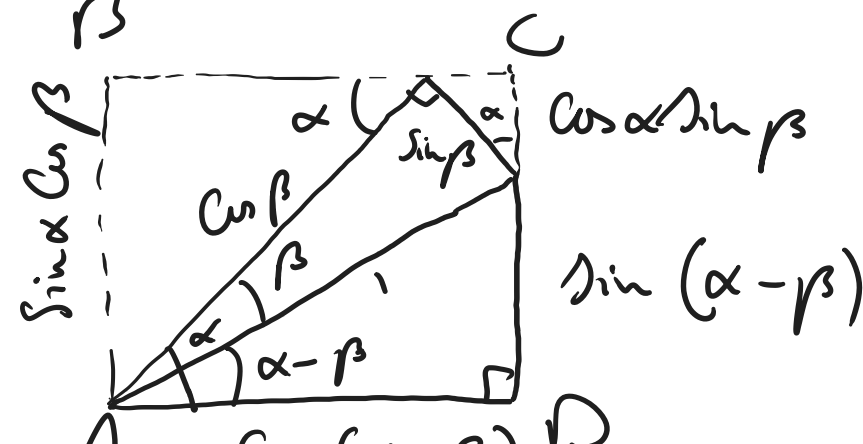
Problem 1)



$$AB = CD \Rightarrow \sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$AB = CD \Rightarrow$$



Problem 2)

a) Prove $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$

$$\vec{a} = a_x \hat{x} + a_y \hat{y} + a_z \hat{z}$$

$$\vec{b} = b_x \hat{x} + b_y \hat{y} + b_z \hat{z}$$

$$\vec{c} = c_x \hat{x} + c_y \hat{y} + c_z \hat{z}$$

$$\begin{aligned} \vec{a} \times (\vec{b} \times \vec{c}) &= \left[a_y (b_x c_y - b_y c_x) - a_z (b_z c_x - b_x c_z) \right] \hat{x} \\ &+ \left[a_z (b_y c_z - b_z c_y) - a_x (b_x c_y - b_y c_x) \right] \hat{y} \\ &+ \left[a_x (b_z c_x - b_x c_z) - a_y (b_y c_z - b_z c_y) \right] \hat{z} \end{aligned}$$

Meanwhile, the RHS is,

$$\begin{aligned} \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b}) &= (a_y c_y b_x + a_z c_z b_x - a_y b_y c_x - a_z c_z c_x) \hat{x} \\ &+ (a_x c_x b_y + a_z c_z b_y - a_x b_x c_y - a_z b_z c_y) \hat{y} \\ &+ (a_x c_x b_z + a_y c_y b_z - a_x b_x c_z - a_y b_y c_z) \hat{z} \end{aligned}$$

Comparing the two expressions, it follows that;

$$\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b}) \quad \checkmark$$

b) Prove $(\vec{a} \times \vec{b}) \times \vec{c} = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{a}(\vec{b} \cdot \vec{c})$

Recall from lecture 2, the anti-commutativity property of the cross product: $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$

$$\Rightarrow (\vec{a} \times \vec{b}) \times \vec{c} = -\vec{c} \times (\vec{a} \times \vec{b})$$

From the BAC-CAB rule proved above, it then follows that

$$(\vec{a} \times \vec{b}) \times \vec{c} = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{a}(\vec{b} \cdot \vec{c}) \quad \checkmark$$

Problem 3) Prove that $\nabla(\log|\vec{r}|) = \frac{\vec{r}}{r^2}$

From lecture notes acm-3, we have in spherical coordinates,

$$\nabla f(r, \theta, \phi) = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\phi}$$

and in Cartesian coordinates,

$$\nabla f = \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z}$$

Given that $f(\vec{r}) = \log(r)$ has spherical symmetry, our life is much easier in spherical coordinates.

$$\Rightarrow \nabla(\log(|\vec{r}|)) = \frac{\partial \log(r)}{\partial r} \hat{r} = \frac{1}{r} \hat{r} = \frac{\vec{r}}{r^2} \quad \checkmark$$

Problem 4) Recall the definition of matrix multiplication from

lecture acm-3. $[AB]_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$ which involves n multiplications.

For square matrices multiplication of rank n , (i, j) indices run from 1 to n , each. Therefore, matrix multiplication of square matrices of rank n , require n^3 scalar multiplications.

Problem 5)

known: m : mass

g : gravity acc.

R : Earth radius.

unknown: v ?

$$[v] = L T^{-1} = [m] [g]^\beta [R]^\gamma = M^\alpha \frac{L^\beta}{T^{2\beta}} L^\gamma$$

$$\Rightarrow \left\{ \begin{array}{l} M^0 = M^\alpha \Rightarrow \alpha = 0 \\ L = L^{\beta+\gamma} \\ T^{-1} = T^{-2\beta} \Rightarrow \beta = \frac{1}{2} \end{array} \right\} \Rightarrow \gamma = \frac{1}{2} \Rightarrow v \propto \sqrt{gR}$$

Problem 6)

$$v(t) = -\frac{g}{\alpha} [1 - \exp(-\alpha t)]$$

$$\alpha t \rightarrow 0: e^{-\alpha t} \approx 1 - \alpha t + \frac{(\alpha t)^2}{2} - \mathcal{O}^3(\alpha t)$$

$$\Rightarrow \lim_{\alpha t \rightarrow 0} v(t) = -\frac{g}{\alpha} [1 - 1 + \alpha t] = -gt$$

where we have ignored the higher-order terms in the Taylor expansion because they do not matter compared to (αt) .

Thus, $v(t) = -gt$ when $\alpha t \rightarrow 0$. This equation makes sense, because at the very beginning, the motion is like a free fall because speed is low and F_D is also low. $\checkmark$

$$\alpha t \rightarrow \infty: \lim_{\alpha t \rightarrow \infty} e^{-\alpha t} = 0 \Rightarrow v(t) = -\frac{g}{\alpha} = \text{constant}$$

This also makes sense because eventually the drag force becomes so strong that completely negates the gravity.

Thus, the object reaches a constant terminal velocity. $\checkmark$

Problem 7)

Consider the limiting extreme cases. If θ is 90° , it takes the projectile forever to reach the cliff.

Thus, the correct answer must yield an infinite time for $\theta = 90^\circ$.

Only the second and third equations satisfy this criterion.

From the geometry of the problem, we also know that the projectile never reaches the cliff if $\theta < 45^\circ$.

$$\text{Only } \frac{1}{\cos \theta} \sqrt{\frac{g L}{2(\tan \theta - 1)}} \text{ satisfies this criterion.}$$

